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Effect of Strategic Memory Advanced Reasoning Training (SMART) therapy for enhancing Final-Year High School Students Career Choices

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Abstract

The SMART program improves students' memory, reasoning, and strategic thinking skills, crucial for academic success and career planning. This study explored the effect of Strategic Memory Advanced Reasoning Training (SMART) for final-year high school students, aiming to enhance their decision-making abilities and prepare them for university. Based on the literature, nine hypotheses were developed with SMART program implementation therapy as an independent variable with four sub-variables: cognitive skills, professional development, social skills, and academic skills, and their impact on the dependent variable, such as career decision-making. Using a smart partial least square-structural equation modeling (PLS-SEM) on 284 high school students, confirmatory factor analysis (CFA) and structural equation modeling (SEM) was implemented to confirm the measurement model. Path analysis was conducted to determine the relationship between independent and dependent variables. Results of the study revealed that SMART therapy significantly enhances cognitive abilities, academic performance, personal development, and social skills, collectively contributing to better career decision-making among final-year high school students. However, the direct impact of SMART on career decision-making was not supported, indicating that additional factors, such as social and emotional influences, play a role. These findings suggest that integrating SMART therapy into high school curricula can better prepare students for future challenges and career opportunities, aligning with Sustainable Development Goal 4 (Quality Education). A collaborative approach among stakeholders, policy support, and innovative practices are recommended to overcome potential obstacles and ensure the successful implementation of SMART therapy in educational settings.

Introduction

High school students require advice and counseling to pursue post-graduation ambitions efficiently (Gamino et al., 2022). As they approach graduation, they face critical decisions about their future academic and career paths, which can be challenging as they must align their career aspirations with their skills and interests (Okolie et al., 2020). The modern school environment necessitates that educators provide instruction while managing various developmental and social challenges, meeting safety and health standards, and adhering to state-mandated learning requirements (Pshembayeva et al., 2022). Although high school educators are state-licensed and capable, the added requirements and complexities within the school environment leave them with little time or ability to guide students effectively toward their chosen ambitions (Llistosella et al., 2023). In Pakistani high schools, this situation is further exacerbated by inadequate guidance from teachers and mentors, leading to confusion and uncertainty about choosing the right major or discipline (Ahmed et al., 2023). Consequently, students often struggle to make informed decisions about their future (Akbar et al., 2023).

In such situations, students face problems regarding their future careers (Mwantimwa, 2021). For example, students with exemplary academic achievements often aim for engineering and medical fields, but average and below-average students are usually unaware of their aptitudes and interests (Rehman et al.,2024). They tend to follow their friends' choices—if their friends get admitted to a business program, they do the same. Some students listen to their parents or family members, while others go to low-budget

colleges and feel stuck, leading to poor decisions about their future. As a result, they may not succeed in their careers and lose interest in their studies (Aslam et al., 2022). In these cases, students need guidance to listen to their inner voices, consider their interests, aptitudes, and skills, and make informed decisions. Therefore, they need help from their teachers and mentors.

The timeliness of decisions is crucial due to the globalized impact of decisions and actions across various settings (Páez-Gallego et al., 2020). Unfortunately, high school students often do not work with vast amounts of data and may be unprepared for life beyond high school (Hsu et al., 2023). Teachers must understand that besides teaching or transferring knowledge, their duty also includes mentoring and guiding students (Keiler et al., 2020; Parker et al., 2021). It involves using visualization tools to help students identify professional, personal, and financial goals to enable student-centered growth as they create and pursue their ambitions (Walker et al., 2022). Expanded duties and responsibilities have shifted educators' focus towards more measurable school outcomes, reducing their ability and available time to mentor students and prepare them for adult life beyond high school graduation (Morrison et al., 2021). Several strategies and approaches have been adopted to facilitate students in such situations (Li et al., 2022). Therapies such as Cognitive Behavioral Therapy (CBT), Mindfulness-Based Stress Reduction (MBSR), Motivational Interviewing (MI), and Career Counseling assist students in analyzing the significance of their researched information, synchronizing their actions and understanding how their ambitions impact other aspects of their lives (Gamino et al., 2022). Through these therapies, students can become more active in articulating a personal vision statement and focusing on their destiny. Additionally, these programs are designed to enhance cognitive skills and support career decision-making (Lewis-Kipkulei et al., 2021).

While these methods provide valuable support in managing stress, improving mental health, and guiding career choices (Dugnol-Menéndez et al., 2021), the Strategic Memory Advanced Reasoning Training (SMART) program offers unique advantages (S. Chapman et al., 2017a). Unlike traditional therapies, SMART focuses on enhancing memory, reasoning, and strategic thinking skills, which are critical for effective learning and decision-making (Gamino et al., 2022). This structured approach not only aids students in making informed career choices but also equips them with advanced cognitive abilities essential for academic success and personal growth (Cook et al., 2014; Cook et al.,2020). This study aims to evaluate the effectiveness of the Strategic Memory Advanced Reasoning Training (SMART) program in helping final-year high school students in Pakistan make informed career choices and improve their readiness for university. Specifically, this study will assess the impact of SMART on several vital areas: Cognitive Abilities, Academic Performance, Career Decision-Making, Personal Development, and Social Skills. These skills are crucial for students' future careers, as they enhance their ability to think strategically, perform academically, make well-informed career decisions, develop personally, and interact socially. The study will provide comprehensive insights into how SMART can support students transitioning from high school to higher education and beyond by focusing on these areas. Although studies on SMART for career choices in high school students are lacking, the evidence of its benefits on cognitive, executive functions, and psychological health suggests that it could indirectly support better

career decision-making processes in this population. The present study would be beneficial in confirming this application.

Overview of previous studies

Although children do not inherently possess executive function skills, they have the innate capacity to develop and enhance these abilities regardless of their home environment (Lehrl et al., 2020). Skills like selective focus and impulse control are crucial for effective learning but are insufficient (Zhao, 2023). As academic and social demands increase, higher-order executive functions such as reasoning, problem-solving, and flexible thinking become indispensable (Rehman et al., 2023). Integrating executive function training into education enhances real-world goal achievement, particularly in academic pursuits (Rehman et al., 2024). The growing evidence of executive function trainability underscores schools' significant responsibility and valuable opportunity when educators recognize their importance (Lau et al., 2021). These skills correlate with higher cognitive functions that support academic success and classroom confidence and extend to fostering more vital social skills and emotional regulation (Wetchasit et al., 2020).

Several studies have explored the efficacy of Strategic Memory Advanced Reasoning Training (SMART) therapy in various educational contexts. A study by Vas et al. (2016) examined the impact of SMART on high school students' cognitive abilities and academic performance. The research, involving a cohort of high school students, revealed significant improvements in working memory, problem-solving skills, and standardized test scores among participants. The study highlighted that student who received SMART training outperformed their peers in critical thinking assessments and exhibited better overall academic performance, suggesting that SMART can effectively enhance cognitive functions necessary for academic success.

Another noteworthy study by Gamino et al. (2022) investigated the role of SMART therapy in career readiness and decision-making among final-year high school students. The longitudinal study followed students who underwent SMART training over six months, assessing their career decision-making skills and university readiness. Results indicated that SMART participants demonstrated superior strategic thinking and planning skills compared to a control group, leading to more informed and confident career choices. Additionally, these students reported higher levels of self-efficacy. They reduced anxiety about their future, underscoring the therapy's potential to support students in navigating the complex transition from high school to higher education and career pathways (Cook et al.,2020). These findings reinforce the importance of incorporating cognitive training programs like SMART into high school curricula to better prepare students for future challenges.

Theoretical framework

The theoretical framework for this study on implementing Strategic Memory Advanced Reasoning Training (SMART) for final-year high school students is grounded in cognitive and educational psychology. This

framework integrates theories on cognitive abilities, academic performance, personal development, social skills, and career decision-making to evaluate how the SMART program can enhance students' readiness for university and their career choices (Chapman et al., 2017b; Chapman et al., 2015; Vas et al., 2016; Zientz et al., 2023). The SMART program is designed to improve cognitive functions such as memory, reasoning, and strategic thinking, which are critical for academic success and career planning (Chapman et al., 2016). Cognitive abilities form the foundation of learning and problem-solving skills, which students need to process information, make informed decisions, and adapt to new situations (Zhong & Xu, 2019). The processing speed theory (Salthouse, 1996; Salthouse & Babcock, 1991) and the intelligence and education framework (Deary et al., 2021) support the importance of enhancing cognitive abilities to achieve better educational outcomes.

Improved cognitive abilities are expected to lead to better academic performance (Chapman et al., 2015). Educational research indicates the relationship between cognitive skills and academic achievements(Aksayli et al., 2019). High academic performance is often associated with effective learning strategies, time management, and applying knowledge to practical scenarios (Peng & Kievit, 2020.). Research by Theobald, (2021) and Yan (2020) emphasizes the role of self-regulated learning and its impact on academic success. The SMART program incorporates activities that foster self-regulation, thereby improving students' academic outcomes (Gati et al., 2019; Suranata et al ., 2019).

As students develop their cognitive and academic skills, they become better equipped to make informed career choices (Gati & Kulcsár, 2021). Career decision-making involves evaluating one's interests, strengths, and market opportunities to select an appropriate career path (R. W. Lent & Brown, 2020). Lent et al. (2019) social cognitive career theory and Super's, (2020) life-span, life-space approach to career development highlight the importance of cognitive and academic skills in making career decisions.

The SMART program also aims to foster personal growth by enhancing self-awareness, goal-setting, and self-efficacy. Personal development is crucial for students to manage stress, stay motivated, and pursue lifelong learning (Fletcher, 2019). Theories of self-determination (Ryan & Deci, 2022) and self-efficacy (Bandura, 1997) provide a basis for understanding how personal development can influence students' career readiness and overall well-being.

Effective communication, teamwork, and empathy are essential social skills that contributing to academic and career success(Paolini, 2020). The SMART program includes activities designed to improve social interactions, which can help students build strong professional networks and navigate social challenges in their academic and career journeys. (Deary et al., 2022) Goleman’s (2006) social intelligence theory and Riggio's et al. (2020) assessment of social skills underscore the significance of these abilities in personal and professional contexts.

Conceptual framework

Globally, teachers are encouraged to adopt theory-driven and research-supported strategies and instructional practices to improve the academic performance of high school students because they have

to move to universities where they continually offer courses, resources, and tidbits to refine and improve skills. Strategic Memory Advanced Reasoning Training (SMART) is a cognitive training program developed by scientists and researchers at the Center for BrainHealth, University of Texas at Dallas. The SMART program teaches strategies to improve critical thinking abilities (Chapman et al., 2017b; Chapman et al., 2015; Vas et al., 2016; Zientz et al., 2023). Multiple randomized trials, including in healthy adults and clinical populations, demonstrated neural, cognitive, and functional gains following SMART (Bailie et al., 2021). The manualized higher-order executive function training program consisted of 10 45-minute classroom sessions delivered over a 1.5-month. The program instructed students in metacognitive strategies and practice that fostered top-down processing via abstracting meanings from information (Gamino et al., 2022). SMART therapy program engaged students in classroom-wide discussions encouraging verbal and written expression of ideas and thought processes facilitated by the instructor's presentation of the program's open-ended questions (Cook et al.,2020).

SMART Therapy and Cognitive Abilities

The SMART program aims to enhance cognitive abilities essential for effective learning and problem-solving (Chapman et al., 2017). Cognitive abilities such as memory, reasoning, and strategic thinking are foundational for academic success and career planning. Research by Salthouse (1996) posits that faster cognitive processing speeds lead to more efficient learning and better academic outcomes. Similarly, Deary et al. (2022) highlighted the importance of intelligence and education, suggesting that enhanced cognitive abilities contribute to higher educational achievements. Vas et al. (2016) found significant improvements in working memory and problem-solving skills among high school students who underwent SMART training. These findings support the hypothesis that SMART therapy positively influences cognitive abilities (see Fig. 1).

H1

SMART positively influences cognitive abilities.

Cognitive Abilities and Career Decision-Making

Cognitive abilities play a crucial role in career decision-making. Lent et al. (2019) social cognitive career theory and Super's (2020) life-span, life-space approach to career development provide a framework for understanding how enhanced cognitive and academic skills aid in making informed career choices. Gamino et al. (2022) demonstrated that students who received SMART training exhibited superior strategic thinking and planning skills, leading to more confident and informed career decisions. These studies suggest that improved cognitive abilities directly impact students' ability to make well-informed career decisions.

H2

Cognitive abilities positively influence career decision-making.

Academic Performance and Career Decision-Making

Academic performance is another critical factor influencing career decision-making. Kuh et al. (2006) and Fokkens-Bruinsma et al. (2021) emphasized the role of self-regulated learning and academic self-efficacy in achieving academic success. These skills help students manage their study time effectively, understand and apply lecture materials, and perform well in exams and assignments, enhancing their ability to make informed career choices. The positive correlation between academic performance and career decision-making is well-supported by empirical evidence (Lent et al.,2020).

H3

Academic performance positively influences career decision-making.

SMART Therapy and Academic Performance

The SMART program's focus on enhancing cognitive abilities naturally extends to improving academic performance (Peng & Kievit, 2020). By fostering skills such as memory, reasoning, and strategic thinking, SMART therapy helps students manage their academic tasks more effectively (Vas et al., 2016, 2020). Vas et al. (2016) found that students who participated in SMART training outperformed their peers in standardized tests and critical thinking assessments. It supports the hypothesis that SMART therapy positively impacts academic performance.

H4

SMART positively influences academic performance.

SMART Therapy and Personal Development

Personal development involves stress management, goal setting, and self-motivation, which are crucial for academic and career success. Ryan and Deci’s (2022) self-determination theory and Bandura's (1997,2006) self-efficacy theory highlight the importance of these aspects in personal growth. SMART therapy includes activities designed to enhance these skills, helping students reflect on their strengths and weaknesses, take responsibility for their development, and pursue lifelong learning(Vas et al.,2020,2016). These theories provide a solid foundation for the hypothesis that SMART therapy positively influences personal development.

H5

SMART positively influences personal development.

SMART Therapy and Social Skills

Effective communication and teamwork are essential for academic and career success. Goleman's (2006) social intelligence theory and Riggio et al. 's (2020) assessment of social skills underscore the significance of these skills. SMART therapy includes components that enhance social skills, such as group activities and discussions. By improving students' ability to communicate and work with others, SMART therapy supports their overall personal and professional development (Goleman et al.,2002).

H6

SMART positively influences social skills.

Social Skills and Career Decision-Making

Social skills are critical in career decision-making, facilitating better networking, communication, and collaboration (Gati et al., 2019). Improved social skills help students seek career advice, participate in group projects, and perform well in interviews, which are all important for making informed career choices (Otto et al., 2019). The positive impact of social skills on career decision-making is supported by theories of social and emotional learning (SEL) (Lent et al., 2017; Rogers et al., 2008).

H7

Social skills positively influence career decision-making.

Personal Development and Career Decision-Making

Personal development, including self-awareness, goal setting, and stress management, is crucial for effective career planning. Ryan and Deci's (2022) and Bandura's (1997) theories suggest that proactive individuals are better equipped to make well-informed career decisions in their personal development. SMART therapy enhances these aspects, supporting students in career planning and decision-making.

H8

Personal development positively influences career decision-making.

By integrating findings from recent studies and theoretical frameworks, this study hypothesizes that the SMART program positively influences cognitive abilities, academic performance, personal development, and social skills, enhancing career decision-making. This framework provides a comprehensive understanding of how the SMART program can enhance students' development, ultimately leading to better career choices and preparedness for university life.

SMART Therapy and Career Decision-Making

The combined influence of enhanced cognitive abilities, improved academic performance, personal development, and social skills facilitated by SMART therapy leads to better career decision-making (Bailie et al., 2021; Laane et al., 2023; Vas et al., 2016). The strategic memory and reasoning skills developed through SMART therapy enable students to make more informed and confident career choices. This

holistic approach aligns with the findings of Gamino et al. (2022) and Vas et al. (2016), who demonstrated the comprehensive benefits of SMART training in preparing students for higher education and career pathways.

H9: SMART positively influences career decision-making.

By integrating findings from recent studies and theoretical frameworks, this study hypothesizes that the SMART program positively influences cognitive abilities, academic performance, personal development, and social skills, enhancing career decision-making (Suranata, 2019).

Methodology

This study employed a quantitative approach with a survey design, utilizing the Partial Least Square-Structural Equation Modeling (PLS-SEM) technique. PLS-SEM is an analytical method to examine complex relationships between variables within a theoretical framework (Hair et al., 2017; Hair et al., 2019). It combines confirmatory factor analysis (CFA) and multiple linear regression, allowing for the simultaneous use of measurement and structural models (Russo & Stol, 2021). To assess these models rigorously, we used the specialized statistical software SmartPLS. PLS-SEM was chosen based on its widespread application and proven effectiveness in existing literature (Zeng et al., 2021). Compared to traditional statistical methods, SEM provides superior results by enhancing the efficiency and rigor of statistical analysis.

Sample

This study was conducted as an experimental study in two government high schools in the Kashmir district. It involved two sections of a girls' high school and two sections of a boys' high school, aiming to enhance students' career choices. A total of 284 students from both high schools participated in this therapy. Purposive sampling was used to collect student data (Campbell et al., 2020). This sample size aligns with guidelines suggesting that empirical research should involve more than 30 but fewer than 500 participants (Lakens, 2022). The demographic profile of the respondents provides a detailed view of the study's target population (see Table 1).

Table 1
Profile of respondents

Demographic	Age	Group	Therapy duration	Frequency (n = 284)	Percentage (%)
Girls' high school A MZF district	15–19	Arts Group	8 weeks	35	12.32%
Girls High School B MZF district	16–20	Science Group	8 weeks	36	12.68%
Boys' high school A MZF district	15–19	Arts Group	8 weeks	32	11.27%
Boys' High School B MZF district	16–20	Science Group	8 weeks	39	13.73%

Questionnaire development

To develop the questionnaire to see the impact of SMART therapy on students' career choice development and preparation for university life, the researcher conducted a thorough literature review and utilized validated constructs from existing research. The questionnaire was designed to measure six key variables: Cognitive Abilities (CA), Academic Performance (AP), Personal Development (PD), Social Skills (SS), Career Decision Making (CDM), and the implementation of the SMART program. Each variable's items were adapted from well-established sources to ensure reliability and validity. The items for Cognitive Abilities were derived from research by Salthouse (1996) and (Deary, 2020), focusing on various cognitive processes such as understanding, retention, application, concentration, problem-solving, and critical analysis. Items for Academic Performance were adapted from studies by Kuh et al. (2006) and (Bandura, 2001) Zimmerman & Schunk (2013), covering aspects such as assignment completion, understanding of lecture materials, exam performance, class participation, and effective study management. The items measuring Personal Development were based on the works of Ryan and Deci (2022) and (Bandura et al., 2003), addressing goal setting, stress management, motivation, personal growth, and proactive learning. For Social Skills, items were taken from Goleman (2006) and Riggio (1986), focusing on making friends, effective communication, social participation, conflict resolution, empathy, and maintaining long-term relationships. Career Decision-making items were sourced from Lent et al. (2019) and Super (2020), which assessed career interests, research, informed choices, goal setting, seeking advice, and confidence in decision-making. The SMART therapy component was informed by Chapman et al. (2015), Gamino et al. (2022), Cook et al. (2014), Cook et al. (2020), and Vas et al. (2015). The questionnaire comprised 40 statements, with each variable represented by five statements. A 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree), was used to evaluate participants' responses. Two statistical tools, SPSS and SmartPLS (PLS-SEM), were used for data analysis. A dataset review was conducted to ensure it met measurement model specifications. Indicator loadings were examined as the first step in the evaluation process. Confirmatory factor analysis (CFA) validated the measurement model, assessing

various types of validity (Hair et al., 2019). Partial least squares structural equation modeling (PLS-SEM) was used to evaluate the proposed model, which was effective with small sample sizes and complex models (Hair et al., 2017). Multivariate statistical analysis using PLS-SEM was conducted in two stages: first, evaluating the measurement model, and then the structural model through SmartPLS 4.0, with 10,000 bootstrap samples to ensure accuracy (Richter et al., 2023)

Therapy implementation

The Strategic Memory Advanced Reasoning Training (SMART) program was implemented over 14 weeks in two government high schools in the Kashmir district to enhance students' career choices. The therapy involved 284 girls' and boys' high school students, with sessions designed to improve cognitive abilities, academic performance, personal development, social skills, and career decision-making. Data for the present study was collected two months after the completion of the training program, right before the university examinations, to assess the training program's impact. The details of the therapy implementation are provided in the table below (see Table 2).

Table 2
Weekly Session Plan for the SMART Program

Week	Focus Area	Activities and Goals	Description
Week 1	Introduction and Memory Techniques	**Introduction to the SMART program, its benefits, and objectives.** - Teaching memory techniques such as mnemonics and visualization. -Interactive memory games are conducted to practice these techniques.	Students were introduced to the SMART program and its benefits. Memory techniques like mnemonics and visualization were taught, followed by interactive memory games.
Week 2	Enhancing Working Memory	**Working Memory Exercises** - Conducting working memory exercises and group activities. - Introducing career exploration tools and resources.	Exercises to enhance working memory were conducted. Group activities to reinforce memory techniques learned in Week 1 were conducted. Tools and resources for career exploration were introduced.
Week 3	Developing Reasoning Skills	**Introduction to Reasoning** - Deductive and inductive reasoning taught - Logic puzzles and brainteasers conducted - Group discussions on problem-solving strategies held	Students were taught deductive and inductive reasoning. Logic puzzles and brainteasers were conducted to develop reasoning skills. Group discussions on various problem-solving strategies were held.
Week 4	Critical Thinking	**Critical Thinking Exercises** - Conducting critical thinking exercises, debates, case studies, and career aptitude tests.	Critical thinking exercises and analysis were conducted. Debates and case studies were used to practice critical thinking. Career aptitude tests and interest inventories were administered.
Week 5	Strategic Planning	**Activities and Goals**: **Goal-setting** - workshops, teaching planning and organizational skills.	Workshops focused on goal setting, planning, and organizing were conducted. Decision-making skills were taught through career-related simulations.

Week	Focus Area	Activities and Goals	Description
		Conducting decision-making simulations related to career choices.	
Week 6	Integration and Application	**Activities and Goals**: - Applying skills to real-world scenarios. - Conducting mock career counseling sessions and providing individual coaching.	Students applied learned skills to real-world scenarios, participated in mock career counseling sessions, and received individual coaching on career planning.
Week 7	Review and Practice	**Activities and Goals**: - Reviewing key concepts and conducting practice sessions with feedback. - Preparing for final assessments.	Key concepts and techniques learned throughout the program were reviewed. Practice sessions with feedback were conducted to reinforce learning. Students were prepared for final assessments.
Week 8	Final Assessment and Feedback	**Activities and Goals**: - Conducting post-training assessments to evaluate the program's impact. - Feedback session and discussion on next steps for students.	Post-training assessments were conducted to evaluate the impact of the program. A feedback session was conducted, and next steps for students were discussed.

Findings of the study

A Confirmatory Factor Analysis (CFA) is the first step to test the reliability and validity of the measurement variables, laying the foundation for subsequent structural analysis. The PLS-SEM is applied to test the hypothesis after proposing the hypothesis or exploring the causal relationships between latent constructs. Internal consistency reliability was assessed using Cronbach's Alpha (CA) and Composite Reliability (CR), as previous studies indicated that CR estimates tend to produce more accurate reliability coefficients (Kalkbrenner, 2023). According to empirical evidence, CA and CR coefficients should always be greater than 0.7 for a good and reliable instrument. For the analysis of the measurement model, Hair and Alamer (2022) suggested evaluating the convergent validity, composite reliability, and the Average Variance Extracted (AVE) value.

Based on the analysis of Table 3 and Fig. 2 provided, the results indicate strong reliability and validity of the constructs used in this study. The table shows that all Cronbach's Alpha (CA) and Composite Reliability (CR) coefficients exceed the recommended threshold of 0.70, demonstrating high internal consistency reliability ((Hamid et al., 2017)Cho, 2016). Specifically, the CA coefficients range from 0.888 to 0.930, and the CR coefficients range from 0.914 to 0.942, indicating reliable measurements across all

constructs. The Average Variance Extracted (AVE) values for each construct also meet the minimum acceptable threshold of 0.50, confirming convergent validity (Hamid et al., 2017). The AVE values in this study range from 0.519 to 0.643. Additionally, the Variance Inflation Factor (VIF) values used to assess multicollinearity were below the critical value of 5, with the highest VIF value being 2.780. It indicates no significant multicollinearity problem in the model (Kim, 2019).

The path analysis model shows that all measurement items have factor loadings higher than 0.50, which is the acceptable threshold for retaining measurement items in the model (Albright & Park, 2009). The factor loadings for the constructs range as follows: Academic Performance (AP) from 0.783 to 0.807, Cognitive Abilities (CA) from 0.701 to 0.758, Career Decision Making (CDM) from 0.761 to 0.840, Personal Development (PD) from 0.760 to 0.835, Smart Program Implementation (SPI) from 0.777 to 0.816, and Social Skills (SS) from 0.759 to 0.798. These loadings indicate strong correlations between the indicators and their respective constructs.

Table 3
Factors loading and assessment of the indicators.

Indicators	Factor Loading	VIF	Cronbach's Alpha	CR	AVE	Source
AP.2	0.807	2.075	0.888	0.914	0.641	Kuh et al. (2006) and Zimmerman & Schunk (2013)
AP.3	0.783	1.915				
AP.4	0.819	2.122				
AP.5	0.800	1.950				
AP.6	0.803	2.063				
AP.7	0.790	1.881				
CA.1	0.710	1.994	0.900	0.915	0.519	Salthouse (1996) and Deary (2020)
CA.10	0.719	2.415				
CA.2	0.732	1.963				
CA.3	0.686	1.896				
CA.4	0.758	2.397				
CA.5	0.718	2.404				
CA.6	0.701	2.020				
CA.7	0.737	2.136				
CA.8	0.720	2.125				
CA.9	0.719	1.971				
CDM.1	0.797	2.270	0.930	0.942	0.643	Lent and Brown (2020) and Super (2020)
CDM.2	0.761	1.970				
CDM.3	0.799	2.317				
CDM.4	0.821	2.528				
CDM.5	0.840	2.780				
CDM.6	0.783	2.152				
CDM.7	0.791	2.155				

(Note: AP: Academic Performance, CA: Cognitive Abilities: Career Decision Making: Personal Developments: Smart Program Implementation Therapy, SS: Social Skills VIFvarianceinflationfactor, CA Cronbachs alpha, CR composite reliability, AVE average variance extracted)

Indicators	Factor Loading	VIF	Cronbach's Alpha	CR	AVE	Source
CDM.8	0.831	2.622				
CDM.9	0.788	2.170				
PD.1	0.781	1.927	0.885	0.913	0.636	Ryan and Deci (2022) and Bandura et al (2023)
PD.2	0.835	2.308				
PD.3	0.788	1.958				
PD.4	0.823	2.303				
PD.5	0.760	1.857				
PD.6	0.795	1.955				
SP1.1	0.777	2.177	0.942	0.950	0.635	Chapman et al. 2015; Gamino et al. 2022; Cook et al. 2014; Cook et al. 2020; Vas et al. 2015
SPI.10	0.797	2.376				
SPI.11	0.786	2.362				
SPI.2	0.778	2.368				
SPI.3	0.790	2.445				
SPI.4	0.794	2.375				
SPI.5	0.806	2.567				
SPI.6	0.816	2.698				
SPI.7	0.813	2.772				
SPI.8	0.801	2.513				
SPI.9	0.801	2.520				
SS.1	0.781	2.127	0.923	0.936	0.618	Goleman (2006) and Riggio (1986)
SS.2	0.798	2.389				
SS.3	0.791	2.123				
SS.4	0.799	2.256				

(Note: AP: Academic Performance, CA: Cognitive Abilities: Career Decision Making: Personal Developments: Smart Program Implementation Therapy, SS: Social Skills VIFvarianceinflationfactor, CA Cronbachs alpha, CR composite reliability, AVE average variance extracted)

Indicators	Factor Loading	VIF	Cronbach's Alpha	CR	AVE	Source
SS.5	0.759	1.972				
SS.6	0.796	2.308				
SS.7	0.787	2.180				
SS.8	0.785	2.130				
SS.9	0.776	2.083				
(Note: AP: Academic Performance, CA: Cognitive Abilities: Career Decision Making: Personal Developments: Smart Program Implementation Therapy, SS: Social Skills VIFvarianceinflationfactor, CA Cronbachs alpha, CR composite reliability, AVE average variance extracted)						

The Heterotrait-Monotrait (HTMT) technique was employed to assess the discriminant validity of the constructs in the model. According to Henseler et al. (2015), an HTMT ratio of less than 0.85 (strict) or less than 0.90 (acceptable) indicates adequate discriminant validity. Table 4 displays the HTMT values for the constructs:

Table 4
Discriminant validity assessment through Heterotrait-Monotrait (HTMT) ratio.

	AP	CA	CDM	PD	SPI	SS
AP						
CA	0.450					
CDM	0.454	0.574				
PD	0.357	0.540	0.575			
SPI	0.466	0.787	0.609	0.561		
SS	0.422	0.598	0.600	0.517	0.702	
(Note: AP: Academic Performance, CA: Cognitive Abilities: Career Decision Making: Personal Developments: Smart Program Implementation Therapy, SS: Social Skills)						

All HTMT values are below the acceptable threshold of 0.90, confirming that discriminant validity is established for all constructs. The Fornell-Larcker criterion was also used to assess discriminant validity. This criterion requires that each construct's square root of the Average Variance Extracted (AVE) be greater than its highest correlation with any other construct. Table 5 presents the Fornell-Larcker criterion values:

Table 5
Discriminant validity through Fornell-Larcker criterion

	AP	CA	CDM	PD	SPI	SS
AP	0.800					
CA	0.418	0.720				
CDM	0.414	0.551	0.802			
PD	0.319	0.495	0.526	0.797		
SPI	0.427	0.785	0.574	0.514	0.797	
SS	0.383	0.578	0.557	0.469	0.658	0.786

(Note: AP: Academic Performance, CA: Cognitive Abilities: Career Decision Making: Personal Developments: Smart Program Implementation Therapy, SS: Social Skills)

The diagonal values in Table 4 represent each construct's square root of the AVE. These values are more significant than the corresponding off-diagonal correlations, further supporting the discriminant validity of the constructs.

After confirming the validity and reliability of the measurement model through the HTMT ratio and Fornell-Larcker criterion, the next stage involved developing the structural model, exploring possible relationships, and testing the hypotheses. By ensuring that both HTMT values and Fornell-Larcker criteria meet the necessary thresholds, the discriminant validity of the constructs in the model is confirmed. This validation process is crucial before proceeding to the structural model analysis, which investigates the causal relationships among the constructs.

Structural Model

Table 6 presents the results of the hypothesis testing through path analysis, including path coefficients (β), T values, P values, effect sizes (f2), and confidence intervals (2.5% CI and 97.5% CI). Each hypothesis was evaluated to determine the impact of various factors on Career Decision Making (CDM) and other related constructs. For H1, it was found that Academic Performance (AP) positively impacts Career Decision Making (CDM) with a significant path coefficient ($\beta = 0.135$, $t = 4.664$, $p < 0.000$) and a small effect size ($f2 = 0.026$). The confidence interval (CI) ranges from 0.078 to 0.190, leading to the acceptance of H1. H2 hypothesized that Cognitive Abilities (CA) would positively impact CDM. However, the path coefficient ($\beta = 0.139$, $t = 1.920$, $p = 0.055$) was not significant, and the effect size ($f2 = 0.013$) was small, with a confidence interval that included zero (-0.002 to 0.282), resulting in the rejection of H2. For H3, Personal Development (PD) was found to have a significant positive impact on CDM ($\beta = 0.240$, $t = 4.523$, $p < 0.000$) with a moderate effect size ($f2 = 0.073$) and a CI ranging from 0.137 to 0.343, leading to the acceptance of H3. H4 posited that Smart Program Implementation Therapy (SPI) would positively impact Academic Performance (AP). It was supported by a significant path coefficient ($\beta = 0.427$, $t = 12.150$, $p <$

0.000), a large effect size (f2 = 0.224), and a CI from 0.360 to 0.496, leading to the acceptance of H4. For H5, SPI was also found to have a significant positive impact on Cognitive Abilities (CA) with a very strong path coefficient (β = 0.785, t = 44.231, p < 0.000), a large effect size (f2 = 1.609), and a CI from 0.752 to 0.821, resulting in the acceptance of H5. H6 hypothesized that SPI would positively impact CDM. However, the path coefficient (β = 0.137, t = 1.946, p = 0.052) was not significant, and the effect size (f2 = 0.011) was small, with a CI including zero (-0.005 to 0.270), leading to the rejection of H6. H7 found that SPI positively impacts Personal Development (PD) with a significant path coefficient (β = 0.514, t = 11.902, p < 0.000), a large effect size (f2 = 0.360), and a CI from 0.431 to 0.598, leading to the acceptance of H7. H8 showed that SPI positively impacts Social Skills (SS) with a significant path coefficient (β = 0.658, t = 19.313, p < 0.000), a large effect size (f2 = 0.763), and a CI from 0.589 to 0.725, leading to the acceptance of H8. Lastly, H9 hypothesized that SS would positively impact CDM. It was supported by a significant path coefficient (β = 0.223, t = 3.820, p < 0.000), a small effect size (f2 = 0.049), and a CI from 0.111 to 0.341, resulting in the acceptance of H9.

Table 6
Hypothesis testing through path analysis.

Hypotheses	Parameter	Path Coefficients (β)	T Values	P values	Effect Size (f2)	2.5% CI	97.5% CI	Decision
H1	AP > CDM	0.135	4.664	0.000	0.026	0.078	0.190	Accepted
H2	CA > CDM	0.139	1.920	0.055	0.013	-0.002	0.282	Rejected
H3	PD > CDM	0.240	4.523	0.000	0.073	0.137	0.343	Accepted
H4	SPI > AP	0.427	12.150	0.000	0.224	0.360	0.496	Accepted
H5	SPI > CA	0.785	44.231	0.000	1.609	0.752	0.821	Accepted
H6	SPI > CDM	0.137	1.946	0.052	0.011	-0.005	0.270	Rejected
H7	SPI > PD	0.514	11.902	0.000	0.360	0.431	0.598	Accepted
H8	SPI > SS	0.658	19.313	0.000	0.763	0.589	0.725	Accepted
H9	SS > CDM	0.223	3.820	0.000	0.049	0.111	0.341	Accepted

(Note: AP: Academic Performance, CA: Cognitive Abilities: Career Decision Making: Personal Developments: Smart Program Implementation Therapy, SS: Social Skills)

The path analysis and model fit assessment for this study are illustrated in Fig. 3 and Table 7, respectively. The structural relationships between the constructs are depicted, showcasing the significance of each hypothesized path. The constructs include Academic Performance (AP), Cognitive Abilities (CA), Career Decision Making (CDM), Personal Development (PD), Smart Program Implementation Therapy (SPI), and Social Skills (SS). The path analysis revealed that AP positively impacts CDM (β = 0.135, t = 4.664, p < 0.000), PD positively impacts CDM (β = 0.240, t = 4.523, p < 0.000), SPI positively impacts AP (β = 0.427, t = 12.150, p < 0.000), CA (β = 0.785, t = 44.231, p < 0.000), PD (β = 0.514, t = 11.902, p < 0.000), and SS (β =

0.658, t = 19.313, p < 0.000), and SS positively impacts CDM (β = 0.223, t = 3.820, p < 0.000). However, CA does not significantly impact CDM (β = 0.139, t = 1.920, p = 0.055), and SPI does not significantly impact CDM (β = 0.137, t = 1.946, p = 0.052). The model's fit was evaluated using several goodness-of-fit indices: Standardized Root Mean Square Residual (SRMR), Chi-square (X^2), and Normed Fit Index (NFI). The results showed SRMR = 0.069, X^2 (Chi-square) = 3478.314, and NFI = 0.915. These indices indicate a good fit of the model, with the SRMR value below the recommended threshold of 0.08 and the NFI value exceeding 0.90. Although the Chi-square value is significant, this is expected with large sample sizes. The model demonstrates adequate fit, supporting most hypotheses, except for the non-significant paths from CA to CDM and SPI to CDM. It suggests that while cognitive abilities and SPI directly affect several constructs, their impact on career decision-making may be mediated by other factors.

Table 7
Goodness of Fit Indices

Fit Indices	Estimated Model
SRMR	0.069
X^2 (Chi-square)	3478.314
NFI	0.915
(Note: SRMR standardized root mean square residual, X2 chi-square, NFI Normed Fit Index, EM estimated. Model)	

Discussion

The present study explored the effectiveness of Strategic Memory Advanced Reasoning Training (SMART) therapy for high school students and its potential to support their transition to university. Nine hypotheses were developed to examine the impact of SMART therapy on enhancing final-year high school students' career choices. The findings revealed significant insights into the benefits of SMART therapy in various educational contexts while highlighting areas where the expected outcomes were not fully realized. SMART therapy has extensively trained teachers and students in diverse scenarios. Previous research has demonstrated that SMART significantly improves gist reasoning, working memory, and executive functions in adults with traumatic brain injury (TBI), highlighting its potential benefits for cognitive control and decision-making processes (Vas et al., 2020).

Additionally, SMART has shown improvements in executive function, memory, and daily functioning, indicating a broad impact on higher-order cognitive skills (Vas et al., 2016). These findings align with our study, where SMART therapy positively influenced cognitive abilities and academic performance. The hypothesis that SMART therapy would positively impact cognitive abilities and career decision-making was supported. The results indicated a strong positive impact of SMART on cognitive abilities, consistent with the processing speed theory (Salthouse, 1996) and the intelligence and education framework (Deary, 2020), emphasizing the importance of cognitive enhancements for better educational outcomes.

However, the hypothesis that cognitive abilities would directly influence career decision-making was not supported. It suggests that while SMART therapy enhances cognitive abilities, these improvements alone may not be sufficient to impact career decision-making directly. This outcome could be attributed to the complexity of career decision-making involving cognitive skills and emotional and social factors, as highlighted by Lent et al. (2019) social cognitive career theory and Super's (2020) life-span, life-space approach to career development. The hypothesis that SMART therapy would positively influence career decision-making through its impact on academic performance was supported, demonstrating that improved academic performance provides a solid foundation for making better career decisions. This finding aligns with research indicating that enhanced academic performance through improved cognitive functions is crucial for effective learning and problem-solving (Donnelly & Lambourne, 2011). Kuh et al. (2006) and Zimmerman & Schunk (2013) also support the idea that self-regulated learning and academic achievements are interlinked, providing a basis for informed career choices. Personal development was another area hypothesized to be positively influenced by SMART therapy, which would enhance career decision-making. The results supported the positive impact of SMART on personal development and the subsequent positive effect of personal development on career decision-making. These findings are consistent with theories of self-determination and self-efficacy, which emphasize the importance of personal growth and self-awareness in managing stress, staying motivated, and making informed career choices (Deci & Ryan, 2000; Bandura, 1997).

The hypothesis that SMART therapy would positively impact social skills was also supported. Improved social skills are crucial for networking, interviewing, and career advancement (Basak et al., 2020). The subsequent hypothesis was confirmed that enhanced social skills would positively influence career decision-making. These results underscore the significance of effective communication, teamwork, and empathy in academic and professional contexts, aligning with Goleman’s (2006) social intelligence theory and Riggio's (1986) social skills assessment. However, the hypothesis that SMART therapy would directly influence career decision-making was not supported. This finding suggests that while SMART therapy enhances several critical skills, the direct pathway from SMART to career decision-making might be mediated by other factors such as cognitive abilities, academic performance, personal development, and social skills. In the Pakistani context, the cultural and educational environment may play a role in this outcome. Students often face external pressures and expectations from family and society regarding their career choices, which can overshadow their cognitive and personal development gains from SMART therapy (Rehman et al., 2024).

By incorporating the findings of this study into the broader context of cognitive and educational psychology, it becomes evident that the SMART therapy program is highly beneficial in enhancing the cognitive and social skills necessary for students' academic and career success. However, the direct impact on career decision-making may require additional support mechanisms, especially in contexts where external pressures significantly influence student choices. The findings of this study show that SMART therapy significantly enhances cognitive abilities, academic performance, personal development, and social skills. These improvements collectively contribute to better career decision-making among final-year high school students. However, further exploration is needed to understand the direct impact of

SMART on career decision-making within Pakistan's cultural and educational context. Integrating cognitive training programs like SMART into high school curricula is essential for preparing students for future challenges and career opportunities. When high school students are well-prepared for their career choices, they are more likely to flourish and be productive in their desired fields at university. This preparedness enhances individual success and contributes to the broader educational and professional landscape. Career development is a crucial focus of the Sustainable Development Goals (SDGs), particularly Goal 4: Quality Education. Goal 4 ensures inclusive and equitable education and promotes lifelong learning opportunities for all. By aligning with this goal, SMART therapy supports the development of essential skills that enable students to make informed career choices, ultimately contributing to sustainable economic growth and personal fulfillment. The emphasis on career readiness within the educational framework helps students transition smoothly from high school to university, fostering an environment where they can achieve their full potential and contribute meaningfully to society.

Conclusion

This study demonstrates the effectiveness of Strategic Memory Advanced Reasoning Training (SMART) in preparing final-year high school students for university and future careers. The findings reveal that SMART therapy significantly enhances cognitive abilities, academic performance, personal development, and social skills, collectively contributing to better career decision-making. By improving these critical areas, SMART therapy helps students make informed career choices and succeed academically. The positive impact of SMART therapy on cognitive abilities aligns with the processing speed theory (Salthouse, 1996) and the intelligence and education framework (Deary & Johnson, 2010), which emphasize the importance of cognitive enhancement for educational success. Improved cognitive abilities lead to better academic performance, as Kuh et al. (2006) and Zimmerman & Schunk (2013) documented. Enhanced academic performance provides a solid foundation for making better career decisions, aligning with the social cognitive career theory (Lent et al.,2019) and Super's (2020) life-span, life-space approach to career development. Personal development, supported by self-determination theories (Ryan & Deci, 2020) and self-efficacy (Bandura et al., 2003), is crucial for managing stress, staying motivated, and making informed career choices. SMART therapy's positive impact on social skills, essential for networking, interviewing, and career advancement, aligns with Goleman’s (2006) social intelligence theory and Riggio's (1986) assessment of social skills. These findings underscore the multifaceted benefits of SMART therapy in preparing students for future challenges. However, the direct influence of cognitive abilities and SMART therapy on career decision-making was not supported, suggesting that social and emotional factors play a significant role in career decision-making. In the Pakistani context, external pressures and expectations from family and society regarding career choices may overshadow the cognitive and personal development gains from SMART therapy (Rehman et al., 2024). It highlights the need for a more holistic approach considering these social factors. To address these findings, it is recommended that educational institutions in Pakistan and similar contexts incorporate SMART therapy into their curricula to better prepare students for future challenges and career opportunities. It aligns with Sustainable Development Goal 4: Quality Education, which promotes lifelong learning opportunities. Integrating cognitive training

programs like SMART can help students flourish in their chosen fields at university and become more productive members of society.

The study has several limitations. The sample size is small, consisting of only four schools from the MZF district of Kashmir. Since Kashmir has ten districts, the findings cannot be generalized nationwide. Future studies should be conducted on a larger scale to include participants and institutions from additional districts or provinces. Additionally, further research should explore the social and emotional factors influencing career decision-making in greater depth. In conclusion, SMART therapy holds promise for enhancing the readiness of high school students for university and future careers by improving cognitive abilities, academic performance, personal development, and social skills. Addressing the limitations and expanding the research can provide a more comprehensive understanding of how to effectively support students in their transition from high school to higher education and beyond.

Declarations

Data Availability Statement:

The data supporting this study's findings are not publicly available due to confidentiality and privacy concerns. The data contain information that could compromise the privacy of research participants. For inquiries about the study, please contact the corresponding author.

Competing interests

The author(s) declared no potential conflicts of interest concerning this article's research, authorship, and publication.

Ethical statement:

This study received ethics approval ZSRT2024203 from the Institutional Ethical Board (IEB). All participants were informed about the purpose of the study and the confidentiality of their personal information. Written informed consent was obtained from each participant before their involvement in the research. Participants were assured that their participation was voluntary and that they had the right to withdraw from the study at any time without any consequences. All data collected for this study were anonymized and securely stored to protect participants' privacy. The research team adhered to the ethical guidelines and principles set forth by Shaanxi Normal University throughout the study to ensure the responsible and respectful treatment of all participants involved.

Consent for Publication

For the present study, informed consent was obtained, and all participants provided written consent. Additionally, the ethics committee reviewed and approved the need for consent.

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Authors' contributions

NR: made significant contributions to the conception and design of the study, writing, data collection, analysis, and interpretation. **XH**: provided valuable insights into the theoretical framework conceptualization, resources, supervision, validation, and proofreading **AM:** Data collection and analysis, writing, review, and editing 4- **NZ** review and editing. All authors read and approved the final manuscript.

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Figures

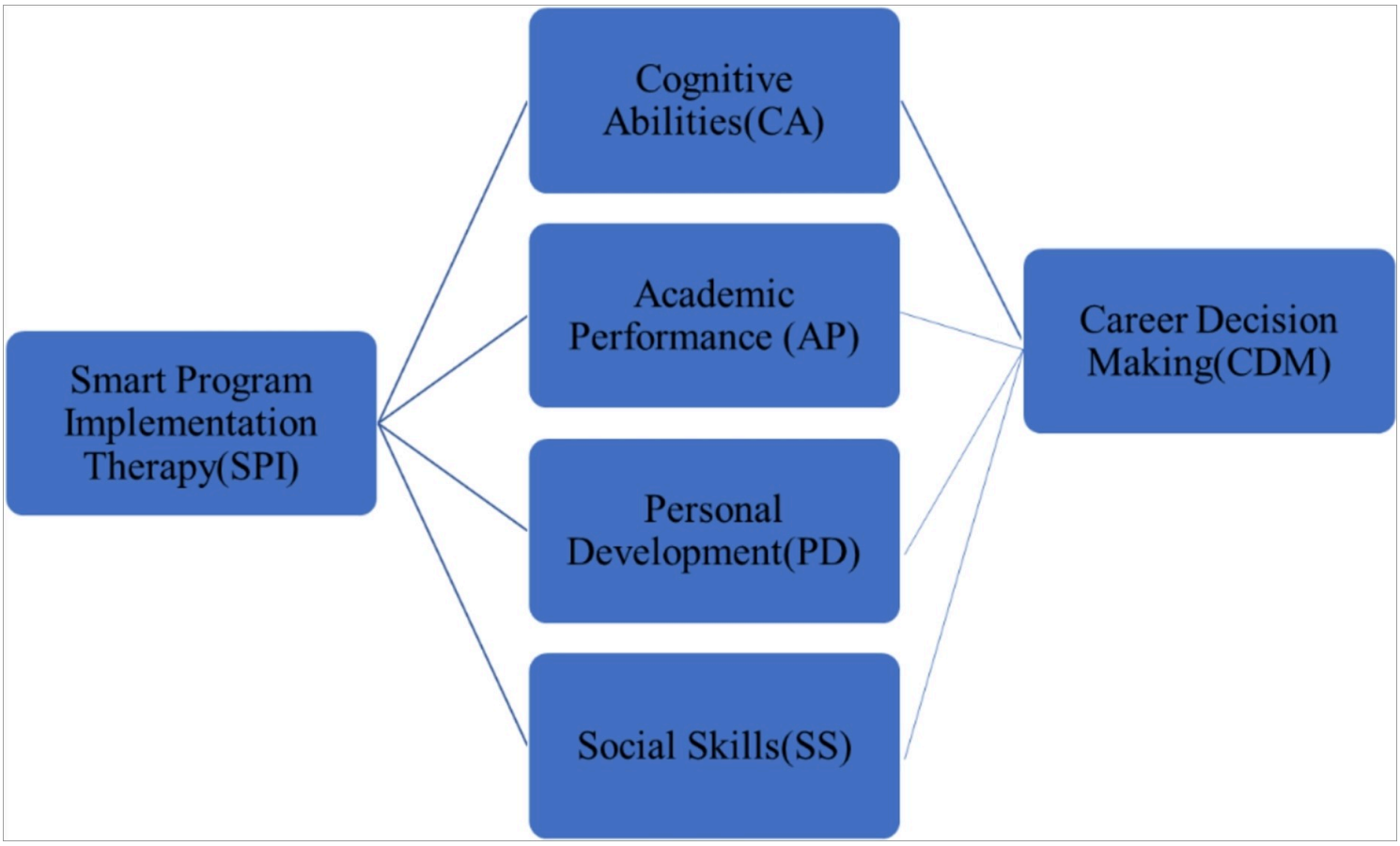


Figure 1

Conceptual framework: The framework shows how SPI influences four key areas: Cognitive Abilities (CA), Academic Performance (AP), Personal Development (PD), and Social Skills (SS). These areas collectively contribute to enhanced Career Decision Making (CDM).

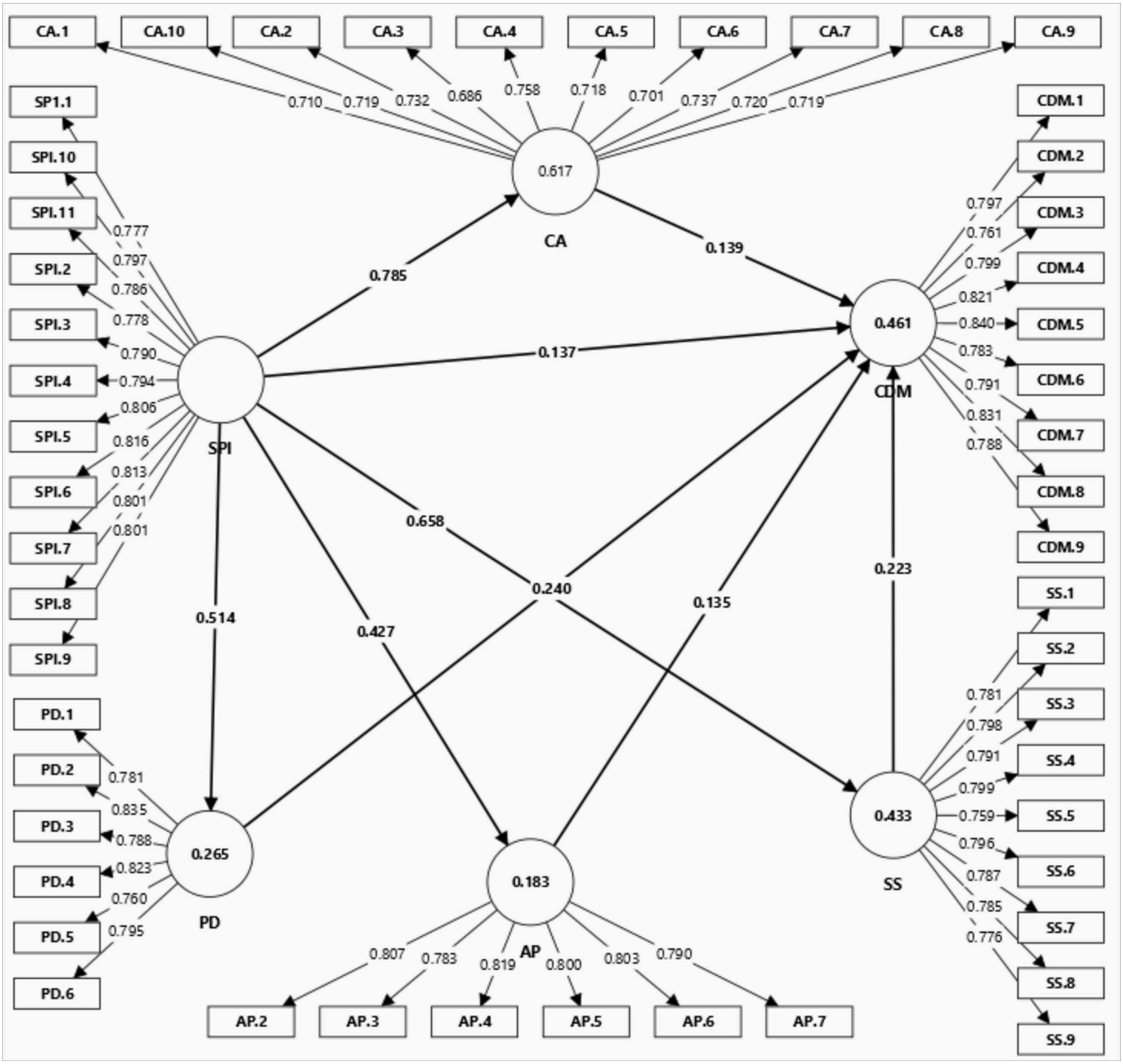


Figure 2

Measurement (path analysis) model.

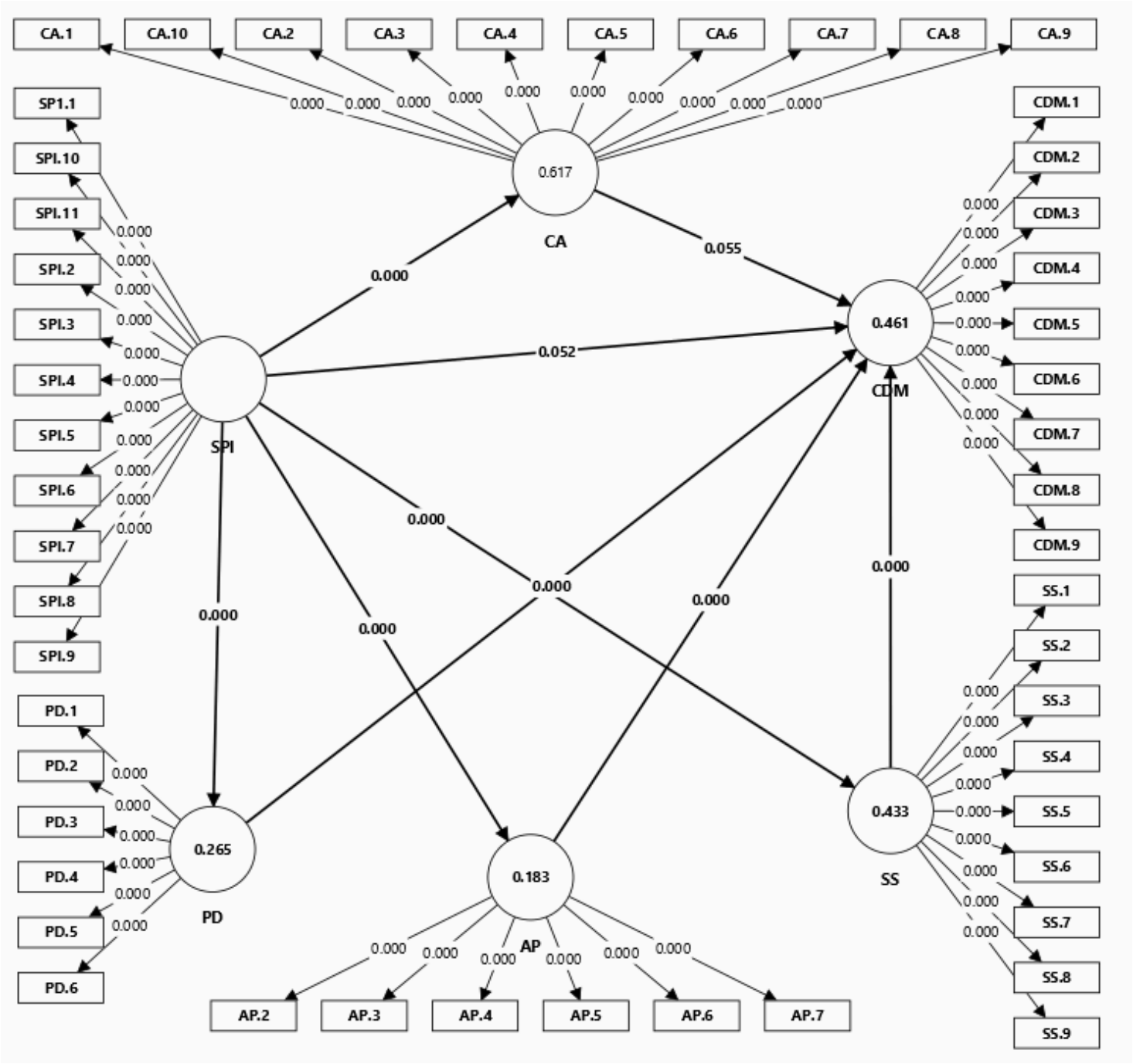


Figure 3

Bootstrap image for path analysis.